

cfbfastR :: CHEAT SHEET

cfbfastR is the SportsDataverse R package for college football: tidy access to the CollegeFootballData.com API (`cfbd_*`), ESPN / Fox / Yahoo live endpoints, full-season play-by-play loaders with expected-points and win-probability model outputs, model calculators, and the PBP parsing / EPA helpers behind them. Python mirror: `sportsdataverse.cfb` in sportsdataverse-py; companions sdvplotR, sportyR, cfbseedR, cfb4th, cfbplotR, recruitR. Page 1 of 2: setup, loaders, models, ESPN live, family map.

v3.1.0

R ≥ 4.1.0 · httr2 · data.table · dplyr
cfbfastR.sportsdataverse.org

GET STARTED

```
# CRAN 3.0.0 · GitHub 3.1.0 (this sheet)
install.packages("cfbfastR")
remotes::install_github(
  "sportsdataverse/cfbfastR")
library(cfbfastR)  # downloads nothing
```

272 exports in five families: `load_cfb_*` / `load_espn_cfb_*` / `load_ncaa_mfb_*` (release data), `cfbd_*` (CollegeFootballData API, key required), `espn_cfb_*`, `fox_cfb_*`, `yahoo_cfb_*` (no key), plus model calculators and PBP / EPA helpers. Every function returns a tibble with a `cfbfastR_type` attribute.

CFBD API KEY

All `cfbd_*` calls need a free key from collegefootballdata.com, stored as `CFBD_API_KEY`. Persist it in `.Renviron`:

```
usethis::edit_r_environ()
# add the line (no quotes), save, restart R
CFBD_API_KEY = YOUR-API-KEY-HERE
# or this session only (with quotes)
Sys.setenv(CFBD_API_KEY = "YOUR-API-KEY")
cfbd_key()            # read it back
has_cfbd_key()        # TRUE / FALSE
cfbd_api_key_info()   # calls left / limit
```

`?register_cfbd` documents the idiom; requests fail fast with a pointer there when the key is missing.

MODEL CALCULATORS

Score any frame (pbp, or a row you type) with the `cfb_model_artifacts` XGBoost models sportsdataverse-py uses, fetched the first time a play is scored:

```
calculate_expected_points(pbp)
calculate_field_goal_probability(
  data.frame(season = 2024,
             yards_to_goal = 25))
calculate_fourth_down(data.frame(
  season = 2024, down = 4, distance = 2,
  yards_to_goal = 45, posteam_total = 52,
  posteam_spread = -3))
cfb_model_card("ep_model")  # features
```

Also `calculate_win_probability` · `_epa` · `_wpa` · `_completion_probability` · `_xpass` · `_two_point_probability` · `_qbr`. Docs: `?calculate_cfb_models`.

DATA LOADERS (RELEASE DATA)

`load_cfb_pbp()` reads the pre-built, EPA/WPA-enriched play-by-play files published to the **sportsdataverse-data** release `cfbfastR_cfb_pbp`. Seasons 2014 to present; no API key.

```
pbp <- load_cfb_pbp(seasons = 2024)
pbp <- load_cfb_pbp(seasons = TRUE) # all
# CFBD-backed (need CFBD_API_KEY), 2001+
sched  <- load_cfb_schedules(2024)
roster <- load_cfb_rosters(2024)
teams  <- load_cfb_teams(fbs_only = TRUE)
# conference membership as of each season
tgs <- load_cfb_team_group_seasons(2024)
```

Loader	Wraps / source
`load_cfb_pbp`	release `cfbfastR_cfb_pbp`
`load_cfb_schedules` · `_rosters` · `_teams`	`cfbd_game_info` · `_team_roster` · `_team_info`
`load_espn_cfb_*` (27) · `load_ncaa_mfb_*` (10)	ESPN + stats.ncaa.org release datasets
`load_cfb_ratings(_weekly)` `_fpi_weekly` `_team_talent` `_recruits` `_returning_production` `*_crosswalk`	ratings · recruiting · id crosswalk releases
`load_cfb_groups` `_group_seasons` `_group_aliases` `load_cfb_team_group_seasons`	release `cfb_groups`: conference / division lineages, 1869+

Progress bars appear automatically when `progressr` is installed.

LOCAL PBP DATABASE

```
# build / refresh a SQLite db of all seasons
update_cfb_db(dbdir = ".",
              dbname = "cfb_pbp_db")
# or hand in your own DBI connection
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "cfb_pbp_db")
update_cfb_db(db_connection = con)
```

Loaders take `dbConnection` / `tablename` to write straight in. Needs `DBI` + `RSQLite`.

PLOT WITH COMPANIONS

```
sportyR::geom_football("ncaa") +
  geom_point(aes(50 - yards_to_goal, 0), pbp)
```

sdvplotR, `league = "cfb"`: `sdv_team_colors()`, `geom_sdv_logos()`, `scale_fill_sdv()`.

ESPN LIVE (NO KEY)

```
espn_cfb_scoreboard(date = 20241130)
espn_cfb_schedule(year = 2024, week = 1,
                  season_type = "regular")
espn_cfb_calendar(year = 2024)
espn_cfb_teams()
espn_cfb_team_roster(team_id = 52,
                     year = 2024)
espn_cfb_team_stats(team_id = 52, year = 2024)
espn_cfb_player_stats(athlete_id = 4426502,
                      year = 2024)
espn_cfb_rankings(year = 2024)
espn_cfb_standings(year = 2024)
```

Play-by-play & game detail

```
espn_cfb_pbp(401628334, epa_wpa = TRUE)
espn_cfb_pbp_v2(401628334, epa_wpa = TRUE,
                output = "lean") # | full
espn_cfb_game_pbp(401628334,
                  participants = "wide")
espn_cfb_game_drives(401628334,
                     plays = "expand")
espn_cfb_game_player_box(401628334)
espn_cfb_game_odds(401628334)
espn_cfb_game_predictor(401628334)
```

With `epa_wpa = TRUE` the v2 engine attaches the pre-game line, so `vegas_wp`, fourth-down and two-point columns fill on ESPN games that have one. Timeouts and 2018+ fair-catch kickoff touchbacks follow sportsdataverse-py.

Ratings & metrics

```
espn_ratings_fpi(year = 2024)
espn_cfb_powerindex(year = 2024)
espn_cfb_qbr(year = 2024)
espn_metrics_wp(game_id = 401628334)
espn_cfb_recruits(year = 2025,
                  max_results = 300)
espn_cfb_clear_cache() # drop memoised
```

More: `espn_cfb_game_*` · `_team_*` · `_player_*` (73 `espn_cfb_*`).

FOX & YAHOO (NO KEY)

```
fox_cfb_pbp(game_id)
fox_cfb_boxscore(game_id)
fox_cfb_team_roster(team_id)
fox_cfb_league_leaders(
  category = "passing", who = "player")
yahoo_cfb_scoreboard(season = 2024,
                     week = 1)
yahoo_cfb_player_season_stats(2024)
```

Also `fox_cfb_odds` · `_standings` · `_team_gamelog`; `yahoo_cfb_boxscore` · `_team_season_stats`.

CFBD_* QUICK MAP (PAGE 2)

Family	Functions
Games	`cfbd_game_info` `_schedule` `_preview` `_box_advanced` `_team_stats` `_player_stats` `_records` `_media` `_weather` `cfbd_calendar`
Drives / plays · live	`cfbd_drives` `cfbd_plays` `cfbd_play_types` `cfbd_play_stats_*` `cfbd_pbp_data` `cfbd_live_*`
Teams	`cfbd_team_info` `_roster` `_talent` `_matchup` `_season_overview` `cfbd_conference_affiliations`
Players / stats	`cfbd_player_info` `_usage` `_returning` `cfbd_stats_season_*` `_game_*` `cfbd_player_season_overview`
Passing · rushing · punting	`cfbd_passing_*` · `_rushing_*` · `_punting_*`: `_players_season` `_players_games` `_teams_season` `_teams_games` `_plays`
Recruiting	`cfbd_recruiting_player` `_team` `_position` `_transfer_portal`
Ratings	`cfbd_ratings_sp` `_srs` `_elo` `_fpi` `cfbd_rankings` `cfbd_playoffs_cfp*`
Betting · metrics	`cfbd_betting_lines` `_ats` · `cfbd_metrics_ppa_*` `_wp` `_wp_pregame` `_fg_ep` `_wepa_*`
Draft / schools	`cfbd_draft_*` `cfbd_coaches` `_profile` `_seasons` `_tenures` `cfbd_venues` `cfbd_conferences`

97 `cfbd_*` exports: all 93 paths of CFBD API v5.33.2.

PYTHON MIRROR

```
from sportsdataverse.cfb import (
    load_cfb_pbp, CFBPlayProcess)
pbp = load_cfb_pbp(seasons=[2024])
proc = CFBPlayProcess(gameId=401628334)
df = proc.run_processing_pipeline()
```

`pip install sportsdataverse`; same EP / WP models, so EPA agrees play for play. py.sportsdataverse.org

COMPANION PACKAGES

cfbseedR	season + 12-team CFP sims · CRAN
cfb4th	fourth-down decision tables · r-univ.
sdvplotR	logos, colours, gt tables · r-univ.
sportyR	college field for ggplot2 · CRAN
cfbplotR	logos / colours for ggplot2 · r-univ.

r-univ.: https://sportsdataverse.r-universe.dev

CollegeFootballData API :: cfbd_* surface & model helpers

All `cfbd_*` need `CFBD_API_KEY` (page 1) · grouped as in the pkgdown reference · Python mirror: **sportsdataverse.cfb** · see also **recruitR** · **cfb4th**

GAMES · DRIVES · PLAYS

GAMES DATA

```
cfbd_game_info(2024, week = 1,
               team = "Georgia",
               quarter_scores = TRUE)
cfbd_game_schedule(2024, week = 1)
cfbd_game_preview(game_id = 401628334)
cfbd_game_team_stats(game_id = 401628334)
cfbd_game_player_stats(2024, week = 1,
                       category = "passing")
cfbd_game_box_advanced(game_id = 401628334)
cfbd_game_records(2024)
cfbd_game_media(2024, week = 1)
cfbd_game_weather(2024, week = 1)
cfbd_calendar(2024)   # week start/end
```

`season_type` : regular / postseason / both · `division` : fbs (default) / fcs / ii / iii. With `game_id` , `year` is optional.

DRIVES DATA

```
cfbd_drives(2024, week = 1, team = "Texas")
cfbd_drives(2024, offense_team = "Oregon",
            defense_conference = "B1G")
```

PLAYS DATA

```
cfbd_plays(2024, week = 1,
           team = "Ohio State")
cfbd_plays(2024, week = 3, play_type = 8)
cfbd_play_types()      # id / text / abbr
cfbd_play_stats_types()
cfbd_play_stats_player(2024, week = 1,
                       team = "Alabama")
cfbd_live_scoreboard(division = "fbs")
cfbd_live_plays(game_id = 401628334)
```

CFBD_PLAYS VS CFBD_PBP_DATA

	`cfbd_plays()`	`cfbd_pbp_data()`
Source	raw CFBD plays	plays + drives + games, cleaned
Modeled	no	`epa_wpa = TRUE` : EP / EPA / WP / WPA + more
Scope	any filter	one `week` (or `team`)

```
cfbd_pbp_data(2024, week = 5,
              epa_wpa = TRUE)
cfbd_pbp_data_v2(2024, week = 5,
                 epa_wpa = TRUE,
                 output = "lean") # | full
```

Whole seasons: `load_cfb_pbp()` (page 1); ESPN twins `espn_cfb_pbp()` / `_v2()` .

CONVENTIONS

`year` is the season start year. Teams take the CFBD school name (`"Ole Miss"`); conferences the abbreviation (`"SEC"` , `"B1G"` , `"B12"`). `excl_garbage_time = TRUE` on stats / PPA / usage.

TEAMS · PLAYERS · STATS · SPLITS

TEAMS DATA

```
cfbd_team_info(conference = "SEC")
cfbd_team_info(only_fbs = FALSE)
cfbd_team_roster(2024, team = "Michigan")
cfbd_team_talent(year = 2024) # 247
cfbd_team_season_overview(2024,
                          team = "Georgia")
cfbd_team_matchup("Auburn", "Alabama",
                  min_year = 2000)
```

PLAYERS DATA

```
cfbd_player_info("Bo Nix", position = "QB")
cfbd_player_usage(2024, team = "Oregon",
                  excl_garbage_time = TRUE)
cfbd_player_returning(2025,
                      conference = "B12")
```

STATS DATA

```
cfbd_stats_season_team(2024,
                       team = "Notre Dame")
cfbd_stats_season_player(2024,
  category = "rushing",
  start_week = 1, end_week = 8)
cfbd_stats_season_advanced(2024,
  excl_garbage_time = TRUE)
cfbd_stats_game_advanced(2024, week = 2,
                         team = "LSU")
cfbd_stats_game_havoc(2024,
                      team = "Penn State")
cfbd_stats_categories() # category=
```

PASSING & RUSHING (2025+)

```
cfbd_passing_players_season(2025,
  team = "Ohio State")
cfbd_passing_plays(2025, week = 5)
cfbd_rushing_teams_season(2025)
```

Splits by pass location and run direction; 0 rows before 2025. `?cfbd_passing` , `?cfbd_rushing` .

PUNTING (2015+)

```
cfbd_punting_players_season(2024,
  team = "Georgia")
cfbd_punting_teams_games(2024, week = 5)
cfbd_punting_plays(2024, week = 5)
```

Each frame has a `punting_*` and a `returning_*` block (`?cfbd_punting`). In all three families divide by the `*_available` columns, not attempts / punts.

RATINGS · BETTING · METRICS · RECRUITS

RANKINGS & RATINGS

```
cfbd_rankings(2024, week = 10) # AP, CFP
cfbd_ratings_sp(2024)          # SP+
cfbd_ratings_sp_conference(2024,
                           conference = "SEC")
cfbd_ratings_srs(2024, team = "Indiana")
cfbd_ratings_elo(2024, week = 8)
cfbd_ratings_fpi(2024)  # FPI via CFBD
```

BETTING DATA

```
cfbd_betting_lines(year = 2024, week = 1,
                   team = "Georgia")
cfbd_betting_lines(game_id = 401628334,
  line_provider = "consensus")
cfbd_betting_ats(2024, conference = "ACC")
```

METRICS DATA (PPA · WP · EPA)

```
cfbd_metrics_ppa_teams(2024,
  excl_garbage_time = TRUE)
cfbd_metrics_ppa_games(2024, week = 1,
                       team = "Miami")
cfbd_metrics_ppa_players_season(2024,
  position = "QB", threshold = 100)
cfbd_metrics_ppa_players_games(2024,
                               week = 1)
cfbd_metrics_ppa_predicted(down = 3,
                           distance = 7)
cfbd_metrics_wp(game_id = 401628334)
cfbd_metrics_wp_pregame(2024, week = 1)
cfbd_metrics_fg_ep()   # FG exp. points
cfbd_metrics_wepa_team_season(2024)
cfbd_metrics_wepa_players_passing(2024,
  team = "Ole Miss")
cfbd_metrics_wepa_players_rushing(2024)
cfbd_metrics_wepa_players_kicking(2024)
```

NFL DRAFT DATA

```
cfbd_draft_picks(2025, college = "Georgia")
cfbd_draft_picks(year = 2024,
                 nfl_team = "Chicago")
cfbd_draft_teams(); cfbd_draft_positions()
```

RECRUITING DATA

```
cfbd_recruiting_player(2025, team = "Texas",
  recruit_type = "HighSchool")
cfbd_recruiting_team(2025)  # class ranks
cfbd_recruiting_position(2020, 2025,
                         team = "Ohio State")
cfbd_recruiting_transfer_portal(2025)
```

`recruit_type` : `"HighSchool"` / `"JUCO"` / `"PrepSchool"` . More in **recruitR**.

SCHOOLS · MODELS · HELPERS

SCHOOLS DATA

```
cfbd_coaches(team = "Georgia",
             min_year = 2016)
cfbd_coaches(last = "Saban")
cfbd_venues();  cfbd_conferences()
```

WHAT EPA_WPA = TRUE ADDS

XGBoost models from the `cfb_model_artifacts` release, fetched on the first scored play (not at `library()`), cached 24 h (falls back to the old `nnet` EP model without `xgboost`):

Columns	Model
`ep_before` `ep_after` `EPA`	EP + era-aware FG
`wp_before` `wpa` · `vegas_wp` `vegas_wpa`	naive · spread-aware WP (2H-kickoff flag fixed)
`cp` `cpoe` · `xpass` `pass_oe`	completion · pass rate
`go_wp` `fg_wp` `punt_wp` `fourth_down_recommendation`	fourth-down decision
`two_pt_wp` `xp_wp` `two_pt_recommendation`	two-point decision

MODEL FUNCTIONS

Function	Role
`calculate_*(df, season)`	10 user-facing scorers (page 1)
`create_qbr(play_df)`	per-QB, per-game QBR table
`create_epa(play_df, ep_model, fg_model, season)`	low-level EP / EPA
`create_wpa_naive(df, wp_model, season)`	low-level WP / WPA
`cfb_model_card` · `cfb_card_features` · `_era_contract` · `_reset_cache`	a model's published card

Also `epa_fg_probs` , `wpa_calcs_naive` , `prep_epa_df_after` .

PBP PARSING HELPERS

`clean_pbp_dat`	play-type flags, scoring
`clean_drive_dat` · `_info`	drive results & ids
`clean_play_text`	normalised `play_text` , read in any case
`add_yardage` · `add_player_cols` · `add_play_counts`	yards (signed `yds_punted`), players, counters
`penalty_detection`	penalty flags, yards



cfbfastR :: CHEAT SHEET

cfbfastR is the SportsDataverse R package for college football: tidy access to the CollegeFootballData.com API (`cfbd_*`), ESPN / Fox / Yahoo live endpoints, full-season play-by-play loaders with expected-points and win-probability model outputs, model calculators, and the PBP parsing / EPA helpers behind them. Python mirror: `sportsdataverse.cfb` in sportsdataverse-py; companions sdvplotR, sportyR, cfbseedR, cfb4th, cfbplotR, recruitR. Page 1 of 2: setup, loaders, models, ESPN live, family map.

v3.1.0

R ≥ 4.1.0 · httr2 · data.table · dplyr
cfbfastR.sportsdataverse.org

GET STARTED

```
# CRAN 3.0.0 · GitHub 3.1.0 (this sheet)
install.packages("cfbfastR")
remotes::install_github(
  "sportsdataverse/cfbfastR")
library(cfbfastR)  # downloads nothing
```

272 exports in five families: `load_cfb_*` / `load_espn_cfb_*` / `load_ncaa_mfb_*` (release data), `cfbd_*` (CollegeFootballData API, key required), `espn_cfb_*`, `fox_cfb_*`, `yahoo_cfb_*` (no key), plus model calculators and PBP / EPA helpers. Every function returns a tibble with a `cfbfastR_type` attribute.

CFBD API KEY

All `cfbd_*` calls need a free key from collegefootballdata.com, stored as `CFBD_API_KEY`. Persist it in `.Renviron`:

```
usethis::edit_r_environ()
# add the line (no quotes), save, restart R
CFBD_API_KEY = YOUR-API-KEY-HERE
# or this session only (with quotes)
Sys.setenv(CFBD_API_KEY = "YOUR-API-KEY")
cfbd_key()            # read it back
has_cfbd_key()        # TRUE / FALSE
cfbd_api_key_info()   # calls left / limit
```

`?register_cfbd` documents the idiom; requests fail fast with a pointer there when the key is missing.

MODEL CALCULATORS

Score any frame (pbp, or a row you type) with the `cfb_model_artifacts` XGBoost models sportsdataverse-py uses, fetched the first time a play is scored:

```
calculate_expected_points(pbp)
calculate_field_goal_probability(
  data.frame(season = 2024,
             yards_to_goal = 25))
calculate_fourth_down(data.frame(
  season = 2024, down = 4, distance = 2,
  yards_to_goal = 45, posteam_total = 52,
  posteam_spread = -3))
cfb_model_card("ep_model")  # features
```

Also `calculate_win_probability` · `_epa` · `_wpa` · `_completion_probability` · `_xpass` · `_two_point_probability` · `_qbr`. Docs: `?calculate_cfb_models`.

DATA LOADERS (RELEASE DATA)

`load_cfb_pbp()` reads the pre-built, EPA/WPA-enriched play-by-play files published to the **sportsdataverse-data** release `cfbfastR_cfb_pbp`. Seasons 2014 to present; no API key.

```
pbp <- load_cfb_pbp(seasons = 2024)
pbp <- load_cfb_pbp(seasons = TRUE) # all
# CFBD-backed (need CFBD_API_KEY), 2001+
sched  <- load_cfb_schedules(2024)
roster <- load_cfb_rosters(2024)
teams  <- load_cfb_teams(fbs_only = TRUE)
# conference membership as of each season
tgs <- load_cfb_team_group_seasons(2024)
```

Loader	Wraps / source
`load_cfb_pbp`	release `cfbfastR_cfb_pbp`
`load_cfb_schedules` · `_rosters` · `_teams`	`cfbd_game_info` · `_team_roster` · `_team_info`
`load_espn_cfb_*` (27) · `load_ncaa_mfb_*` (10)	ESPN + stats.ncaa.org release datasets
`load_cfb_ratings(_weekly)` `_fpi_weekly` `_team_talent` `_recruits` `_returning_production` `*_crosswalk`	ratings · recruiting · id crosswalk releases
`load_cfb_groups` `_group_seasons` `_group_aliases` `load_cfb_team_group_seasons`	release `cfb_groups`: conference / division lineages, 1869+

Progress bars appear automatically when `progressr` is installed.

LOCAL PBP DATABASE

```
# build / refresh a SQLite db of all seasons
update_cfb_db(dbdir = ".",
              dbname = "cfb_pbp_db")
# or hand in your own DBI connection
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "cfb_pbp_db")
update_cfb_db(db_connection = con)
```

Loaders take `dbConnection` / `tablename` to write straight in. Needs `DBI` + `RSQLite`.

PLOT WITH COMPANIONS

```
sportyR::geom_football("ncaa") +
  geom_point(aes(50 - yards_to_goal, 0), pbp)
```

sdvplotR, `league = "cfb"`: `sdv_team_colors()`, `geom_sdv_logos()`, `scale_fill_sdv()`.

ESPN LIVE (NO KEY)

```
espn_cfb_scoreboard(date = 20241130)
espn_cfb_schedule(year = 2024, week = 1,
                  season_type = "regular")
espn_cfb_calendar(year = 2024)
espn_cfb_teams()
espn_cfb_team_roster(team_id = 52,
                     year = 2024)
espn_cfb_team_stats(team_id = 52, year = 2024)
espn_cfb_player_stats(athlete_id = 4426502,
                      year = 2024)
espn_cfb_rankings(year = 2024)
espn_cfb_standings(year = 2024)
```

Play-by-play & game detail

```
espn_cfb_pbp(401628334, epa_wpa = TRUE)
espn_cfb_pbp_v2(401628334, epa_wpa = TRUE,
                output = "lean") # | full
espn_cfb_game_pbp(401628334,
                  participants = "wide")
espn_cfb_game_drives(401628334,
                     plays = "expand")
espn_cfb_game_player_box(401628334)
espn_cfb_game_odds(401628334)
espn_cfb_game_predictor(401628334)
```

With `epa_wpa = TRUE` the v2 engine attaches the pre-game line, so `vegas_wp`, fourth-down and two-point columns fill on ESPN games that have one. Timeouts and 2018+ fair-catch kickoff touchbacks follow sportsdataverse-py.

Ratings & metrics

```
espn_ratings_fpi(year = 2024)
espn_cfb_powerindex(year = 2024)
espn_cfb_qbr(year = 2024)
espn_metrics_wp(game_id = 401628334)
espn_cfb_recruits(year = 2025,
                  max_results = 300)
espn_cfb_clear_cache() # drop memoised
```

More: `espn_cfb_game_*` · `_team_*` · `_player_*` (73 `espn_cfb_*`).

FOX & YAHOO (NO KEY)

```
fox_cfb_pbp(game_id)
fox_cfb_boxscore(game_id)
fox_cfb_team_roster(team_id)
fox_cfb_league_leaders(
  category = "passing", who = "player")
yahoo_cfb_scoreboard(season = 2024,
                     week = 1)
yahoo_cfb_player_season_stats(2024)
```

Also `fox_cfb_odds` · `_standings` · `_team_gamelog`; `yahoo_cfb_boxscore` · `_team_season_stats`.

CFBD_* QUICK MAP (PAGE 2)

Family	Functions
Games	`cfbd_game_info` `_schedule` `_preview` `_box_advanced` `_team_stats` `_player_stats` `_records` `_media` `_weather` `cfbd_calendar`
Drives / plays · live	`cfbd_drives` `cfbd_plays` `cfbd_play_types` `cfbd_play_stats_*` `cfbd_pbp_data` `cfbd_live_*`
Teams	`cfbd_team_info` `_roster` `_talent` `_matchup` `_season_overview` `cfbd_conference_affiliations`
Players / stats	`cfbd_player_info` `_usage` `_returning` `cfbd_stats_season_*` `_game_*` `cfbd_player_season_overview`
Passing · rushing · punting	`cfbd_passing_*` · `_rushing_*` · `_punting_*`: `_players_season` `_players_games` `_teams_season` `_teams_games` `_plays`
Recruiting	`cfbd_recruiting_player` `_team` `_position` `_transfer_portal`
Ratings	`cfbd_ratings_sp` `_srs` `_elo` `_fpi` `cfbd_rankings` `cfbd_playoffs_cfp*`
Betting · metrics	`cfbd_betting_lines` `_ats` · `cfbd_metrics_ppa_*` `_wp` `_wp_pregame` `_fg_ep` `_wepa_*`
Draft / schools	`cfbd_draft_*` `cfbd_coaches` `_profile` `_seasons` `_tenures` `cfbd_venues` `cfbd_conferences`

97 `cfbd_*` exports: all 93 paths of CFBD API v5.33.2.

PYTHON MIRROR

```
from sportsdataverse.cfb import (
    load_cfb_pbp, CFBPlayProcess)
pbp = load_cfb_pbp(seasons=[2024])
proc = CFBPlayProcess(gameId=401628334)
df = proc.run_processing_pipeline()
```

`pip install sportsdataverse`; same EP / WP models, so EPA agrees play for play. py.sportsdataverse.org

COMPANION PACKAGES

cfbseedR	season + 12-team CFP sims · CRAN
cfb4th	fourth-down decision tables · r-univ.
sdvplotR	logos, colours, gt tables · r-univ.
sportyR	college field for ggplot2 · CRAN
cfbplotR	logos / colours for ggplot2 · r-univ.

r-univ.: https://sportsdataverse.r-universe.dev

CollegeFootballData API :: cfbd_* surface & model helpers

All `cfbd_*` need `CFBD_API_KEY` (page 1) · grouped as in the pkgdown reference · Python mirror: **sportsdataverse.cfb** · see also **recruitR** · **cfb4th**

GAMES · DRIVES · PLAYS

GAMES DATA

```
cfbd_game_info(2024, week = 1,
               team = "Georgia",
               quarter_scores = TRUE)
cfbd_game_schedule(2024, week = 1)
cfbd_game_preview(game_id = 401628334)
cfbd_game_team_stats(game_id = 401628334)
cfbd_game_player_stats(2024, week = 1,
                       category = "passing")
cfbd_game_box_advanced(game_id = 401628334)
cfbd_game_records(2024)
cfbd_game_media(2024, week = 1)
cfbd_game_weather(2024, week = 1)
cfbd_calendar(2024)   # week start/end
```

`season_type` : regular / postseason / both · `division` : fbs (default) / fcs / ii / iii. With `game_id` , `year` is optional.

DRIVES DATA

```
cfbd_drives(2024, week = 1, team = "Texas")
cfbd_drives(2024, offense_team = "Oregon",
            defense_conference = "B1G")
```

PLAYS DATA

```
cfbd_plays(2024, week = 1,
           team = "Ohio State")
cfbd_plays(2024, week = 3, play_type = 8)
cfbd_play_types()      # id / text / abbr
cfbd_play_stats_types()
cfbd_play_stats_player(2024, week = 1,
                       team = "Alabama")
cfbd_live_scoreboard(division = "fbs")
cfbd_live_plays(game_id = 401628334)
```

CFBD_PLAYS VS CFBD_PBP_DATA

	`cfbd_plays()`	`cfbd_pbp_data()`
Source	raw CFBD plays	plays + drives + games, cleaned
Modeled	no	`epa_wpa = TRUE` : EP / EPA / WP / WPA + more
Scope	any filter	one `week` (or `team`)

```
cfbd_pbp_data(2024, week = 5,
              epa_wpa = TRUE)
cfbd_pbp_data_v2(2024, week = 5,
                 epa_wpa = TRUE,
                 output = "lean") # | full
```

Whole seasons: `load_cfb_pbp()` (page 1); ESPN twins `espn_cfb_pbp()` / `_v2()` .

CONVENTIONS

`year` is the season start year. Teams take the CFBD school name (`"Ole Miss"`); conferences the abbreviation (`"SEC"` , `"B1G"` , `"B12"`). `excl_garbage_time = TRUE` on stats / PPA / usage.

TEAMS · PLAYERS · STATS · SPLITS

TEAMS DATA

```
cfbd_team_info(conference = "SEC")
cfbd_team_info(only_fbs = FALSE)
cfbd_team_roster(2024, team = "Michigan")
cfbd_team_talent(year = 2024) # 247
cfbd_team_season_overview(2024,
                          team = "Georgia")
cfbd_team_matchup("Auburn", "Alabama",
                  min_year = 2000)
```

PLAYERS DATA

```
cfbd_player_info("Bo Nix", position = "QB")
cfbd_player_usage(2024, team = "Oregon",
                  excl_garbage_time = TRUE)
cfbd_player_returning(2025,
                      conference = "B12")
```

STATS DATA

```
cfbd_stats_season_team(2024,
                       team = "Notre Dame")
cfbd_stats_season_player(2024,
  category = "rushing",
  start_week = 1, end_week = 8)
cfbd_stats_season_advanced(2024,
  excl_garbage_time = TRUE)
cfbd_stats_game_advanced(2024, week = 2,
                         team = "LSU")
cfbd_stats_game_havoc(2024,
                      team = "Penn State")
cfbd_stats_categories() # category=
```

PASSING & RUSHING (2025+)

```
cfbd_passing_players_season(2025,
  team = "Ohio State")
cfbd_passing_plays(2025, week = 5)
cfbd_rushing_teams_season(2025)
```

Splits by pass location and run direction; 0 rows before 2025. `?cfbd_passing` , `?cfbd_rushing` .

PUNTING (2015+)

```
cfbd_punting_players_season(2024,
  team = "Georgia")
cfbd_punting_teams_games(2024, week = 5)
cfbd_punting_plays(2024, week = 5)
```

Each frame has a `punting_*` and a `returning_*` block (`?cfbd_punting`). In all three families divide by the `*_available` columns, not attempts / punts.

RATINGS · BETTING · METRICS · RECRUITS

RANKINGS & RATINGS

```
cfbd_rankings(2024, week = 10) # AP, CFP
cfbd_ratings_sp(2024)          # SP+
cfbd_ratings_sp_conference(2024,
                           conference = "SEC")
cfbd_ratings_srs(2024, team = "Indiana")
cfbd_ratings_elo(2024, week = 8)
cfbd_ratings_fpi(2024)  # FPI via CFBD
```

BETTING DATA

```
cfbd_betting_lines(year = 2024, week = 1,
                   team = "Georgia")
cfbd_betting_lines(game_id = 401628334,
  line_provider = "consensus")
cfbd_betting_ats(2024, conference = "ACC")
```

METRICS DATA (PPA · WP · EPA)

```
cfbd_metrics_ppa_teams(2024,
  excl_garbage_time = TRUE)
cfbd_metrics_ppa_games(2024, week = 1,
                       team = "Miami")
cfbd_metrics_ppa_players_season(2024,
  position = "QB", threshold = 100)
cfbd_metrics_ppa_players_games(2024,
                               week = 1)
cfbd_metrics_ppa_predicted(down = 3,
                           distance = 7)
cfbd_metrics_wp(game_id = 401628334)
cfbd_metrics_wp_pregame(2024, week = 1)
cfbd_metrics_fg_ep()   # FG exp. points
cfbd_metrics_wepa_team_season(2024)
cfbd_metrics_wepa_players_passing(2024,
  team = "Ole Miss")
cfbd_metrics_wepa_players_rushing(2024)
cfbd_metrics_wepa_players_kicking(2024)
```

NFL DRAFT DATA

```
cfbd_draft_picks(2025, college = "Georgia")
cfbd_draft_picks(year = 2024,
                 nfl_team = "Chicago")
cfbd_draft_teams(); cfbd_draft_positions()
```

RECRUITING DATA

```
cfbd_recruiting_player(2025, team = "Texas",
  recruit_type = "HighSchool")
cfbd_recruiting_team(2025)  # class ranks
cfbd_recruiting_position(2020, 2025,
                         team = "Ohio State")
cfbd_recruiting_transfer_portal(2025)
```

`recruit_type` : `"HighSchool"` / `"JUCO"` / `"PrepSchool"` . More in **recruitR**.

SCHOOLS · MODELS · HELPERS

SCHOOLS DATA

```
cfbd_coaches(team = "Georgia",
             min_year = 2016)
cfbd_coaches(last = "Saban")
cfbd_venues();  cfbd_conferences()
```

WHAT EPA_WPA = TRUE ADDS

XGBoost models from the `cfb_model_artifacts` release, fetched on the first scored play (not at `library()`), cached 24 h (falls back to the old `nnet` EP model without `xgboost`):

Columns	Model
`ep_before` `ep_after` `EPA`	EP + era-aware FG
`wp_before` `wpa` · `vegas_wp` `vegas_wpa`	naive · spread-aware WP (2H-kickoff flag fixed)
`cp` `cpoe` · `xpass` `pass_oe`	completion · pass rate
`go_wp` `fg_wp` `punt_wp` `fourth_down_recommendation`	fourth-down decision
`two_pt_wp` `xp_wp` `two_pt_recommendation`	two-point decision

MODEL FUNCTIONS

Function	Role
`calculate_*(df, season)`	10 user-facing scorers (page 1)
`create_qbr(play_df)`	per-QB, per-game QBR table
`create_epa(play_df, ep_model, fg_model, season)`	low-level EP / EPA
`create_wpa_naive(df, wp_model, season)`	low-level WP / WPA
`cfb_model_card` · `cfb_card_features` · `_era_contract` · `_reset_cache`	a model's published card

Also `epa_fg_probs` , `wpa_calcs_naive` , `prep_epa_df_after` .

PBP PARSING HELPERS

`clean_pbp_dat`	play-type flags, scoring
`clean_drive_dat` · `_info`	drive results & ids
`clean_play_text`	normalised `play_text` , read in any case
`add_yardage` · `add_player_cols` · `add_play_counts`	yards (signed `yds_punted`), players, counters
`penalty_detection`	penalty flags, yards